

Week 5: DOM Manipulation & Events

1. Selecting Elements, Manipulating DOM

The DOM (Document Object Model) represents the structure of an HTML document. JavaScript can be used to select and manipulate DOM elements such as headings, buttons, and input fields. Developers can change text, styles, attributes, and content dynamically. This helps create interactive and user-friendly web applications.

2. Event Handling (Click, Submit, Input, etc.)

Event handling is used to respond to user actions like clicking a button, submitting a form, or typing in an input field. JavaScript detects these actions and performs specific tasks based on user interaction. This improves functionality and makes web applications dynamic.

3. Event Listeners, Event Delegation

Event listeners are methods used to attach actions to HTML elements when events occur. For example, clicking a button can trigger a function. Event delegation is a technique where a parent element handles events for its child elements, making code more efficient and reducing memory usage. These concepts improve performance in larger applications.

4. Forms Validation using JS

Form validation ensures that user input is correct before data is submitted. JavaScript can check required fields, email format, password length, and other input rules. This prevents invalid data from being processed and improves user experience and application security.

Conclusion

This week focused on DOM manipulation and event handling in JavaScript. By learning how to select elements, handle events, use event listeners, and validate forms, a strong understanding of interactive frontend development was developed. These skills are essential for creating responsive and user-friendly web applications.